ĐẠI HỌC HOA SEN

TÔN TRỌNG SỰ KHÁC BIỆT

WEEK 6: BUSINESS PROBLEMS, INFORMATION NEEDS & REQUIREMENTS

Course: Business Systems Analysis

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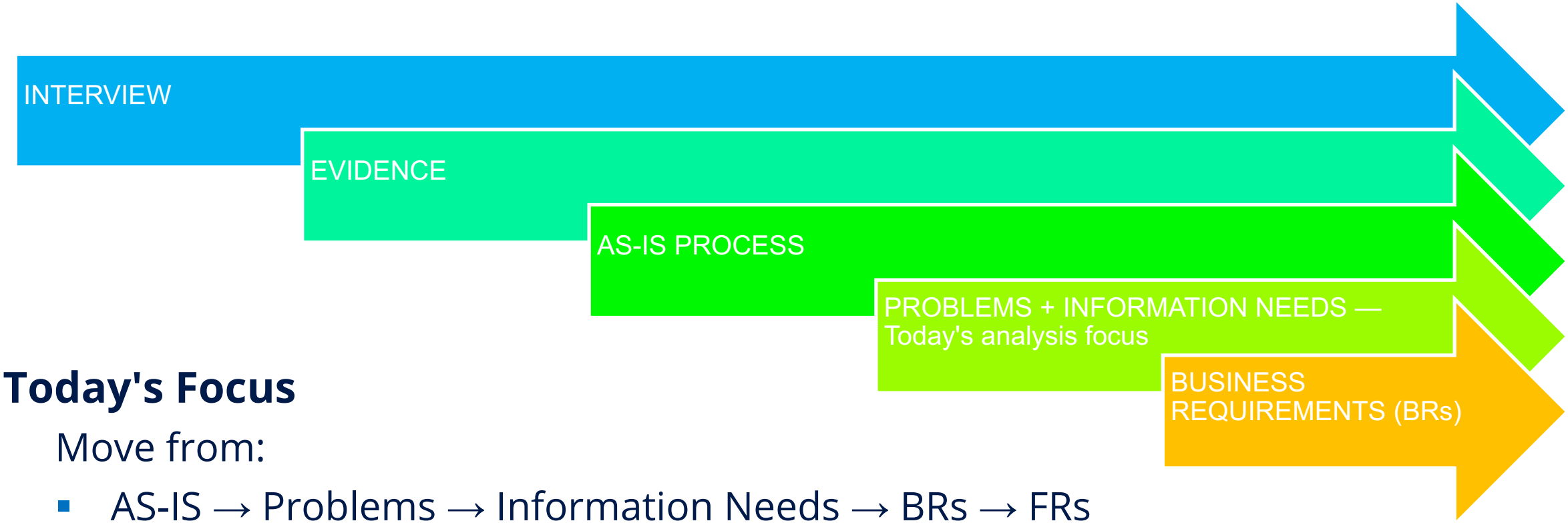
Faculty: Faculty of Technology

Learning Focus

By the end of this session, you should be able to:

- Identify a business problem from an AS-IS process and supporting evidence.
- Identify the stakeholders affected and their information needs.
- Distinguish Business Requirements (BRs) from Functional Requirements (FRs).
- Write clear, testable requirements linked to business needs.
- Establish traceability between evidence, problems, information needs, and requirements.
- Identify candidate Odoo applications that may support the business process at a later project stage.

Where Are We in the Project?



Today's Focus

Move from:

- AS-IS → Problems → Information Needs → BRs → FRs

The approved business process packages help you maintain a coherent project scope and prepare for the later Odoo solution stage.

How We Will Work Today

- Phase 1 (10 min): Review the approved process package and current AS-IS
- Phase 2 (15 min): Mini-lesson — Swimlane modeling fundamentals & Convert AS-IS flowchart to swimlane model
- Phase 3 (15 min): Identify an evidence-supported business problem
- Phase 4 (15 min): Identify affected stakeholders and information needs
- Phase 5 (15 min): Learn the difference between BRs and FRs
- Phase 6 (30 min): Draft BRs and FRs
- Phase 7 (15 min): Review traceability and peer-review requirements
- Phase 8 (15 min): Individual quick quiz and reflection

Swimlane Modeling Fundamentals

Why Swimlanes?

Simple flowcharts show WHAT happens, but Swimlane models explicitly show WHO is responsible for each action and HOW data/documents cross departmental boundaries.

4 Core BPMN Swimlane Building Blocks:

1. Swimlanes (Pools & Lanes):
   - Pool: The overall organization or business scenario.
   - Lanes: Internal departments or roles (e.g., Sales Rep, Warehouse Clerk, Accountant).
2. Flow Objects:
   - Start Event (Thin Circle): The exact event that triggers the process (e.g., Customer Quote Request).
   - Activity / Task (Rounded Rectangle): Action verb + Noun (e.g., Verify Credit Limit).
   - End Event (Thick Circle): Final business state (e.g., Invoice Payment Reconciled).
3. Gateways (Diamond Shapes):
   - Exclusive Gateway (XOR): Decision point with mutually exclusive paths (e.g., Stock Available? Yes / No).
4. Connecting Objects:
   - Sequence Flow (Solid Line): Sequence of activities within the same lane.
   - Data Object / Message Flow (Dashed Line with Document Icon): Information handoffs across lanes (e.g., Purchase Order, Invoice).

Golden Rule: Every process step MUST live inside a designated lane. An unassigned step creates operational ambiguity.

Convert AS-IS Flowchart to Swimlane Model

Task Instructions for Groups (15 Minutes):

Take your Week 5 simple flowchart and convert it into a standardized AS-IS Swimlane Diagram on Miro.

Step-by-Step Conversion Guidelines:

Step 1: Identify Your Lanes

- Create 3–4 horizontal or vertical lanes based on your role evidence (e.g., Customer, Sales, Warehouse, Finance).

Step 2: Map Triggers & Handoffs

- Place the Start Event in the lane of the actor who initiates the process (e.g., Customer sending a Quote Request).
- Move activity boxes into the specific lane of the role executing that action.

Step 3: Define Decision Gateways & Document Flows

- Replace ambiguous decision points with Exclusive Gateways (e.g., Check Material Stock -> Approved/Rejected).
- Draw Data Objects (e.g., Form A, Sales Order, Invoice) along the arrows when information moves from one lane to another.

Step 4: Highlight Operational Friction (Bottlenecks)

- Mark handoff delays, missing data fields, or duplicated checks with a red warning icon [!] for your Requirements Traceability Matrix.

Group Deliverable: A completed AS-IS Swimlane Diagram ready for friction analysis and TO-BE redesign.

How the Approved Process Packages Work

Each business scenario has five recommended business processes, allowing each group member to take responsibility for one process.

Group Guidelines

- Use the approved package for your registered business scenario.
- Assign one primary process to each group member.
- Use evidence to validate the proposed process scope, activities, and actors.
- Prioritize at least three core processes for detailed analysis in the Phase 1 project review.
- Keep each process separately scoped. Do not combine five different processes into one oversized diagram.

Important

- The approved process list is a starting point for project organization, not proof that every activity has already been confirmed.
- If evidence is incomplete, identify what needs further investigation instead of inventing process facts.

Process Metadata: Define Every Process Clearly

Use the following structure for each process.

Metadata	Definition
Process Name	Use the [Verb] + [Noun] format, such as Checking Product Availability.
Trigger	The exact event that starts the process.
End Point	The final state when the process is completed.
Main Actors	The roles or departments involved.

Example — Checking Product Availability

Process Name: Checking Product Availability

Trigger: Sales requests a stock check for products in a customer order.

End Point: The availability status is returned to Sales.

Main Actors: Sales Staff, Inventory Clerk.

Scope Rule

A process should have a clear trigger and a specific business outcome. It does not automatically need to include invoicing and payment.

Scenario 1: Fashion & Apparel Chain

Process Name	Trigger	End Point	Main Actors
BP1. Processing Customer Orders	A customer submits an order for clothing or accessories.	The order is confirmed, rejected, or cancelled, and its status is recorded.	Customer, Sales Staff, Warehouse Clerk
BP2. Checking Product Availability	Sales requests a stock check for selected products, sizes, or colors.	Availability or shortage information is returned to Sales.	Sales Staff, Inventory Clerk
BP3. Reviewing Replenishment Needs	Product stock reaches the defined reorder threshold.	The replenishment request is approved, deferred, or recorded for follow-up.	Inventory Controller, Purchasing Staff, Manager
BP4. Processing Purchase Orders	An approved replenishment request is received.	The purchase order is sent to the supplier and confirmation is recorded.	Purchasing Staff, Supplier, Approving Manager
BP5. Receiving Purchased Goods	A supplier delivers goods against a purchase order.	Received quantities and discrepancies are recorded; stock information is updated or flagged.	Warehouse Clerk, Purchasing Staff, Supplier

Scenario 2: Consumer Electronics Store

Process Name	Trigger	End Point	Main Actors
BP1. Processing Customer Orders	A customer submits an order for electronic products.	The order decision and status are recorded.	Customer, Sales Staff, Warehouse Clerk
BP2. Checking Product Availability	Sales requests confirmation of product model or variant availability.	Available quantities and shortages are reported to Sales.	Sales Staff, Inventory Clerk
BP3. Processing Warranty Claims	A customer submits a warranty claim with product and serial number information.	The claim decision and next action are recorded and communicated.	Customer, Customer Service Staff, Technician
BP4. Processing Product Returns	A customer submits a return or exchange request.	The decision and resulting exchange, refund, or rejection are recorded.	Customer, Customer Service Staff, Warehouse Clerk, Finance Staff
BP5. Replenishing Replacement Parts	A required replacement part is unavailable or below the defined stock level.	The part is replenished, received, and recorded as available for service use.	Technician, Parts Inventory Clerk, Purchasing Staff, Supplier

Scenario 3: Home Decor & Custom Furniture

Process Name	Trigger	End Point	Main Actors
BP1. Processing Custom Furniture Orders	A customer submits an order for customized furniture.	The order decision and agreed specifications are recorded.	Customer, Sales Consultant
BP2. Validating Custom Specifications	A custom furniture request includes dimensions, materials, or design requirements.	Specifications are confirmed as feasible or returned for clarification.	Customer, Sales Consultant, Design/Technical Staff
BP3. Planning Manufacturing Orders	A confirmed custom furniture order requires production planning.	The manufacturing order, required materials, and production schedule are recorded.	Production Planner, Production Manager, Inventory Controller
BP4. Replenishing Raw Materials	Production planning identifies a shortage of required materials.	Required materials are procured, received, and recorded as available for production.	Purchasing Staff, Supplier, Warehouse Clerk, Production Planner
BP5. Processing Delivery Orders	Finished furniture is marked ready for dispatch.	Delivery arrangements and delivery status are recorded.	Warehouse Staff, Delivery Coordinator, Customer

Scenario 4: Cosmetics & Beauty Chain

Process Name	Trigger	End Point	Main Actors
BP1. Processing Customer Orders	A customer submits an order for cosmetics or beauty products.	The order decision and status are recorded.	Customer, Sales Staff, Warehouse Clerk
BP2. Checking Batch and Expiry Information	An order requires batch or expiry verification.	An eligible batch is selected, or a stock/expiry issue is reported.	Warehouse Clerk, Inventory Controller
BP3. Processing Customer Returns	A customer submits a return or exchange request.	The return decision and resulting refund, exchange, or rejection are recorded.	Customer, Customer Service Staff, Warehouse Clerk, Finance Staff
BP4. Replenishing Product Inventory	Product stock reaches its reorder threshold or a shortage is identified.	Replenishment is completed and received quantities are recorded.	Inventory Controller, Purchasing Staff, Supplier, Warehouse Clerk
BP5. Recording Expired or Damaged Products	A product is identified as expired, damaged, or unsuitable for sale.	The product status and corresponding stock adjustment or disposal decision are recorded.	Warehouse Clerk, Inventory Controller, Manager

Scenario 5: FMCG Wholesale Distribution

Process Name	Trigger	End Point	Main Actors
BP1. Processing B2B Customer Orders	A wholesale customer submits an order for specified quantities.	The order is confirmed, revised, or rejected, and its status is recorded.	Customer/Buyer, Sales Executive
BP2. Checking Bulk Stock Availability	Sales requests a stock check for a bulk order.	Full, partial, or unavailable quantities are reported to Sales.	Sales Executive, Warehouse Supervisor
BP3. Handling Order Shortages	The stock check identifies insufficient quantities to fulfill the order as requested.	The customer's decision on partial fulfillment, backorder, substitution, or cancellation is recorded.	Sales Executive, Customer/Buyer, Warehouse Supervisor
BP4. Scheduling Customer Deliveries	A confirmed order requires delivery planning.	A delivery slot or schedule is assigned and communicated.	Logistics Coordinator, Warehouse Supervisor, Sales Executive, Customer
BP5. Replenishing Inventory	Stock falls below the required level or a shortage requires replenishment.	Replenishment is completed and received quantities are recorded as available stock.	Inventory Controller, Purchasing Staff, Supplier, Warehouse Clerk

Scenario 6: Office Supplies & Stationery

Process Name	Trigger	End Point	Main Actors
BP1. Processing Corporate Orders	A corporate customer submits an order for office supplies.	The order decision and status are recorded.	Corporate Customer, Sales Account Executive
BP2. Verifying Contract Prices	A corporate order references an agreed contract, price list, or discount arrangement.	Applicable pricing is verified and recorded in the quotation or order.	Sales Account Executive, Contract/Account Manager
BP3. Checking Product Availability	A proposed or confirmed order requires stock verification.	Available quantities and shortages are reported.	Sales Staff, Warehouse Clerk
BP4. Processing Purchase Orders	An approved purchase request is available for required supplies.	The purchase order is sent to the supplier and confirmation is recorded.	Purchasing Staff, Supplier, Approving Manager
BP5. Receiving Purchased Goods	A supplier delivers office supplies against a purchase order.	Received quantities and discrepancies are recorded and inventory information is updated.	Warehouse Clerk, Purchasing Staff, Supplier

Scenario 7: Industrial Spare Parts

Process Name	Trigger	End Point	Main Actors
BP1. Processing Spare-Part Quotation Requests	A factory customer requests a quotation for a specified spare part.	A quotation with price, availability, and expected lead time is issued, or the request is declined.	Customer/Purchasing Agent, Technical Sales Engineer
BP2. Matching Spare-Part Requests	A part number or technical specification is received.	A suitable part is identified, or clarification is requested.	Technical Sales Engineer, Customer, Inventory Controller
BP3. Checking Spare-Part Availability	A matched spare part is required for a customer request or quotation.	Available quantity and any shortage are confirmed and reported.	Inventory Controller, Technical Sales Engineer
BP4. Procuring Spare Parts	A required spare part is unavailable and procurement is authorized.	A purchase order is placed and supplier availability and lead-time information are recorded.	Purchasing Staff, Supplier, Technical Sales Engineer
BP5. Receiving Purchased Spare Parts	A supplier delivers spare parts against a purchase order.	Part identity and quantities are verified, receipt is recorded, and discrepancies are flagged.	Warehouse Clerk, Inventory Controller, Supplier

Scenario 8: Custom Gift Hamper Assembly

Process Name	Trigger	End Point	Main Actors
BP1. Processing Custom Hamper Orders	A customer submits an order for a customized gift hamper.	The order decision and agreed requirements are recorded.	Customer, Sales Representative
BP2. Confirming Hamper Specifications	A customer requests custom contents, packaging, ribbons, or branded details.	Specifications and required quantities are approved or returned for clarification.	Customer, Sales Representative, Production/Packaging Staff
BP3. Checking Component Availability	An approved hamper specification requires component availability confirmation.	Component quantities and shortages are confirmed and reported.	Inventory Clerk, Production/Packaging Staff
BP4. Processing Assembly Orders	A confirmed order has sufficient information to initiate assembly.	Required hampers are assembled and completed quantities and status are recorded.	Production Planner, Assembly Staff, Warehouse Clerk
BP5. Replenishing Packaging Materials	Packaging materials fall below the required level or are insufficient for a confirmed order.	Required materials are procured, received, and recorded as available.	Inventory Controller, Purchasing Staff, Supplier, Warehouse Clerk

Scenario 9: Bicycle & Scooter Assembly

Process Name	Trigger	End Point	Main Actors
BP1. Processing Custom Bicycle Orders	A customer submits an order for a configured bicycle or scooter.	The order and selected components are confirmed or returned for revision.	Customer, Sales Representative
BP2. Validating Component Compatibility	A customer-selected component configuration is submitted for technical review.	The configuration is validated, or a compatibility issue and required correction are recorded.	Sales Representative, Assembly Mechanic, Customer
BP3. Checking Component Availability	A validated configuration requires component availability confirmation.	Available and missing components are identified and reported.	Inventory Controller, Assembly Mechanic
BP4. Processing Assembly Work Orders	A confirmed assembly order is released with validated specifications.	Assembly is completed, or blocking issues are recorded and the work-order status is updated.	Production Planner, Bicycle Mechanic, Warehouse Clerk
BP5. Replenishing Bicycle Components	Required components are unavailable or fall below the defined stock level.	Components are procured, received, and recorded as available for assembly.	Inventory Controller, Purchasing Staff, Supplier, Warehouse Clerk

Scenario 10: Coffee Roasting & Packaging

Process Name	Trigger	End Point	Main Actors
BP1. Processing Finished-Goods Orders	A customer submits an order for packaged coffee products.	The order decision and status are recorded.	Customer, Sales Staff, Warehouse Clerk
BP2. Planning Coffee Production Batches	Customer demand or stock levels indicate a need for additional production.	The planned batch quantity, schedule, and raw material requirements are recorded.	Production Planner, Inventory Controller, Production Manager
BP3. Replenishing Raw Coffee Beans	The production plan identifies insufficient raw coffee bean inventory.	Required raw beans are procured, received, and recorded as available for production.	Purchasing Staff, Supplier, Warehouse Clerk, Production Planner
BP4. Processing Coffee Production Orders	A production batch is approved and released for roasting and packaging.	Packaged coffee quantities are recorded and finished goods become available in inventory.	Production Planner, Roasting Operator, Packaging Staff, Warehouse Clerk
BP5. Recording Batch Yield and Quality Results	A roasting batch is completed and its actual output must be recorded.	Input/output quantities, yield, and relevant quality results are recorded against the batch.	Roasting Operator, Production Supervisor, Quality Staff

REQUIREMENTS ANALYSIS

Keep Each Process Specific

A business function is broader than an individual business process.

Too Broad	More Specific
Sales Management	Processing Customer Orders
Inventory Management	Checking Product Availability
Purchasing	Processing Purchase Orders
Warehouse Management	Receiving Purchased Goods
Production Management	Processing Assembly Orders

Process Scope Questions

- What event triggers the process?
- What specific transaction are we following?
- What happens from start to finish?
- What is the final business outcome?
- Who are the main actors?

Important

A process should have its own clear boundary. Connected processes may provide inputs to each other, but they should not automatically be combined into one large process.

Start With Evidence

Use your group's revised AS-IS process and the interview evidence collected in previous weeks.
Look for observations such as:

- delays;
- missing or inaccurate information;
- unclear responsibilities;
- repeated data entry;
- unnecessary waiting;
- stock or order discrepancies;
- unclear decisions or exception handling.

Ask These Questions

- What happened?
- What evidence supports this observation?
- Which stakeholder is affected?
- What information does that stakeholder need?
- Why does this matter to the business?

Do not assume that every manual activity is a business problem. Identify why it matters.

Distinguish Observation, Problem, Cause, and Impact

- These are related but different concepts.

Concept	Meaning	Example
Observation	What was observed in the current process	Sales sends Warehouse a message to check stock.
Business Problem	What is not working effectively	Sales cannot reliably access stock availability information when handling an order.
Possible Cause	A factor that may contribute to the problem	Stock information may be maintained separately.
Business Impact	The effect on the process or organization	Order confirmation may be delayed.

- **Key Rule**
  - An observation is not automatically a confirmed root cause.
  - Use evidence to support conclusions, and mark possible causes as uncertain when they have not been verified.

Identify Stakeholder Information Needs

An information need describes information a stakeholder requires to perform a task, coordinate work, monitor a situation, or make a decision.

Example: Stock Availability

Stakeholder	Information Need
Sales Staff	Current product availability before confirming an order
Warehouse Staff	Accurate product and stock information
Purchasing Staff	Information about items requiring replenishment
Manager	Visibility of stock shortages and operational status
Customer	Reliable information about product availability

Your Task

- For your selected process, identify:
- who is affected;
- what information they need;
- when they need it;
- why they need it.

Do not write a system feature yet.

From AS-IS to Information Need

AS-IS Evidence

- Sales receives a customer order and sends a message to Warehouse to check stock availability.
- Warehouse checks the stock information and replies.
- Sales sometimes has to wait before confirming the order.

Step 1 — Observation

- Sales depends on Warehouse to obtain product availability information.

Step 2 — Business Problem

- Sales cannot reliably access stock availability information when handling an order.

Step 3 — Affected Stakeholders

- Sales Staff
- Warehouse Staff
- Customer

Step 4 — Information Need

- Sales needs timely and reliable product availability information before confirming an order.

Important

- Do not turn an information need into a technology recommendation too early.

Business Requirement vs. Functional Requirement vs. Solution

These represent different levels of specification.

Level	Question	Example
Business Requirement (BR)	What does the business need to achieve or support?	The business needs order confirmation to be based on reliable product availability information.
Functional Requirement (FR)	What should the system do to support that need?	The system shall display available product quantities to authorized Sales Staff during order preparation.
Solution	What system or configuration could provide the capability?	Evaluate the relevant Odoo Sales and Inventory capabilities in the configured environment.

Remember

- A BR describes a business need or desired outcome.
- An FR specifies a system behavior.
- A solution describes a proposed implementation approach.

This week, draft BRs and FRs. Do not finalize the Odoo solution yet.

Writing High-Quality Business Requirements

A good BR should:

- describe a genuine business need;
- connect to an identified problem or business objective;
- focus on a business outcome or capability;
- be clear and understandable;
- avoid prescribing a technical solution too early.

Weak BR

- The company needs Odoo Inventory.
- This names a product rather than explaining the business need.

Better BR

- The business needs Sales Staff to verify product availability before confirming customer orders.

Quick Test

- Ask:
  - Why does the business need this?
- If the answer is unclear, revisit the problem and its supporting evidence.

Writing Testable Functional Requirements

A functional requirement describes what the system must do to support an identified business need.

Recommended Format

- The system shall + specific action + object + condition/context.

Weak FR

- The system should improve inventory management.
- This is vague and difficult to test.

Better FRs

- FR-01: The system shall display the available quantity of each selected product to authorized Sales Staff during sales order preparation.
- FR-02: The system shall warn Sales Staff when the requested quantity exceeds the available quantity.

A Good FR Should Be

- clear and specific;
- focused on one main behavior;
- testable;
- linked to an identified business need and process.

Avoid vague statements such as “The system shall be user-friendly” unless the expected behavior can be defined and verified.

Check Requirement Quality

- Review each requirement before adding it to your project.

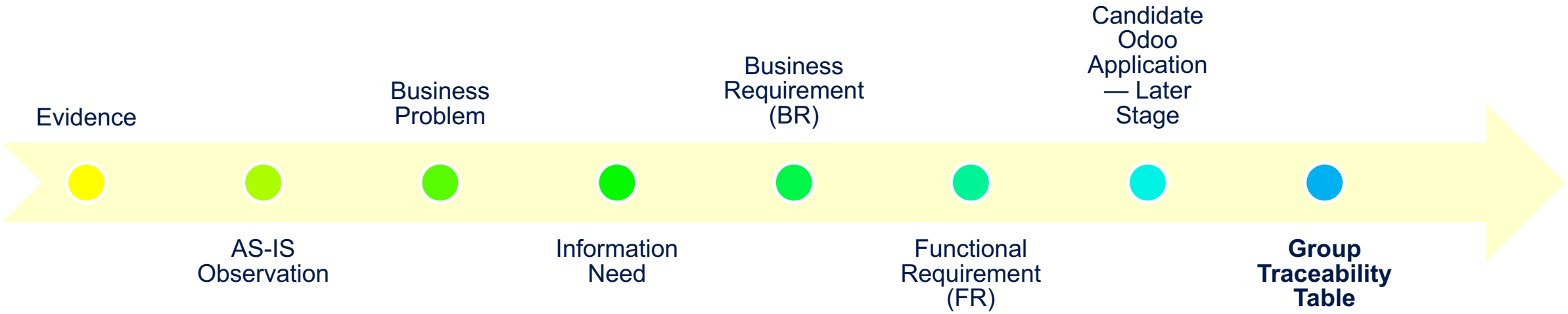
Check	Question
Business relevance	What problem or information need does it address?
Clarity	Can different readers understand the same meaning?
Specificity	Does it describe one main need or behavior?
Testability	Can we determine whether it has been satisfied?
Traceability	Can we trace it back to evidence and the related process?
Scope	Does it belong to the assigned business scenario?
Solution neutrality	Does it avoid prescribing technology too early?

- If a requirement cannot be justified by evidence or a valid business need, review it before keeping it.

Requirements Traceability

Requirements should not appear without an explanation of where they came from.

Traceability Chain



Requirements Traceability

Group Traceability Table

Field	Group's Analysis
Evidence ID / Source	
AS-IS Observation	
Business Problem	
Affected Stakeholder	
Information Need	
BR ID	
FR ID(s)	
Related Business Process	
Candidate Odoo App	

Every FR should be linked to a business need and related process. Do not invent a parent business requirement simply to complete the table.

Activity 1: Identify a Business Problem

Use your group's revised AS-IS process and supporting interview evidence.

- **Tasks**
- Select one process that your group has evidence to analyze.
- Identify one important observation from the current process.
- State the business problem supported by that observation.
- Identify the stakeholder affected.
- Identify the stakeholder's information need.
- Record any unresolved question.

Activity 1: Identify a Business Problem

Group Worksheet

Item	Group's Answer
Business Scenario	
Selected Process	
AS-IS Evidence	
Business Problem	
Affected Stakeholder	
Information Need	
Unresolved Question	

Rule: If the evidence does not support a conclusion, mark it as uncertain rather than inventing facts.

Activity 2: Draft BRs and FRs

Use the business problem and information need identified in Activity 1.

Tasks

1. Write 1–2 Business Requirements.
2. Write 3–4 Functional Requirements that support those needs.
3. Link each requirement to the relevant process.
4. Check that every FR is clear and testable.
5. Identify candidate Odoo applications without finalizing the solution.

Requirements Table

ID	Requirement	Related Problem / Need	Related Process
BR-01			
BR-02			
FR-01			
FR-02			

Example

BR-01: The business needs Sales Staff to verify product availability before confirming customer orders.
FR-01: The system shall display the available quantity of each selected product during sales order preparation.
FR-02: The system shall warn Sales Staff when the requested quantity exceeds the available quantity.
Do not copy the example blindly. Requirements must be based on your group's selected process and evidence.

Peer Review: Are the Requirements Good?

Exchange your requirements table with another group.

- **Business Requirement Check**
  - ❑ The BR responds to an identifiable business problem or need.
  - ❑ It describes a business outcome or capability.
  - ❑ It does not simply name a software product.
  - ❑ It is clear and understandable.
- **Functional Requirement Check**
  - ❑ The FR describes a specific system behavior.
  - ❑ The FR is clear enough to test.
  - ❑ The FR is linked to a business requirement or information need.
  - ❑ The FR is related to a process in the assigned scenario.
  - ❑ The FR does not prematurely prescribe a technical implementation.

Give Two Comments

One requirement that is clear and well justified: …

One requirement that needs revision, with a reason: …

The authors should use the feedback to revise their requirements.

Week 6 Deliverables

Group Working Output

By the end of Week 6, your group should have:

- A confirmed package of five related business processes for your assigned scenario.
- A clear process assignment for each group member.
- One selected process linked to the revised AS-IS model.
- Evidence supporting one important business problem.
- An identified stakeholder and information need.
- Preliminary Business Requirements.
- Preliminary Functional Requirements.
- A traceability table connecting evidence, needs and requirements.
- Candidate Odoo applications for later consideration.
- Keep these outputs on your Miro board or in your project working document.

Important

- The process package organizes the five members' responsibilities. The Phase 1 report should prioritize at least three core processes for deeper analysis in line with the course project requirements.

Roadmap to Phase 1 and Final Project

Week	Project Development
W1	Understand the enterprise and business scope
W2	Interview stakeholders and collect evidence
W3	Draft AS-IS processes
W4	Experience business processes through simulation
W5	Review evidence and refine AS-IS
W6	Identify problems, information needs, BRs and FRs
W7–W8	Refine process analysis and requirements; prepare Phase 1 report
W9	Phase 1 presentation and review
W10–W13	Use cases, prototyping and solution specification
W14–W15	Final project presentations and defense

Week 9 Focus

Present your enterprise context, stakeholder analysis, core business processes, AS-IS/TO-BE analysis and requirements in accordance with the published project requirements.

Detailed Use Case Specifications and solution demonstration belong to the later project stage.